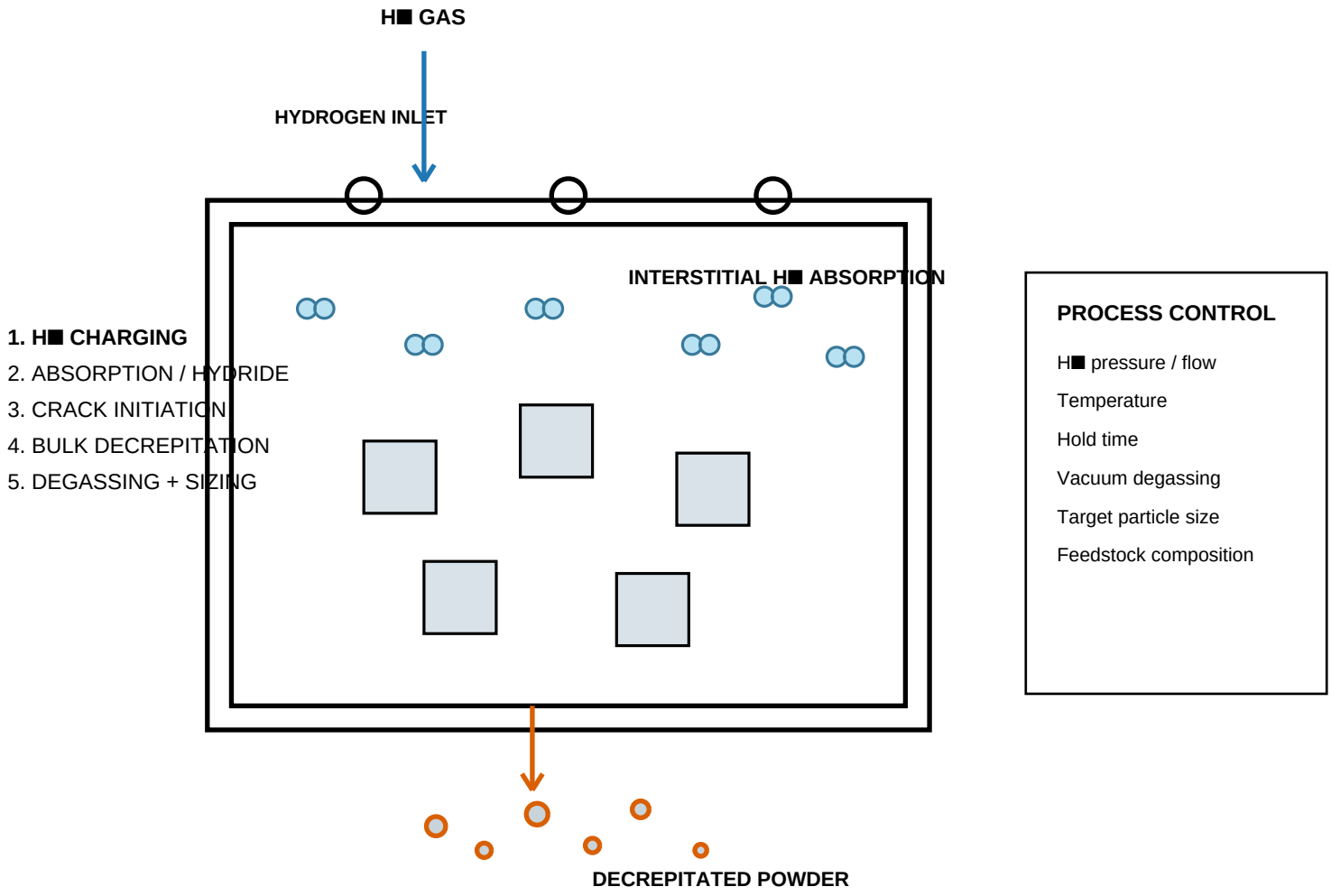
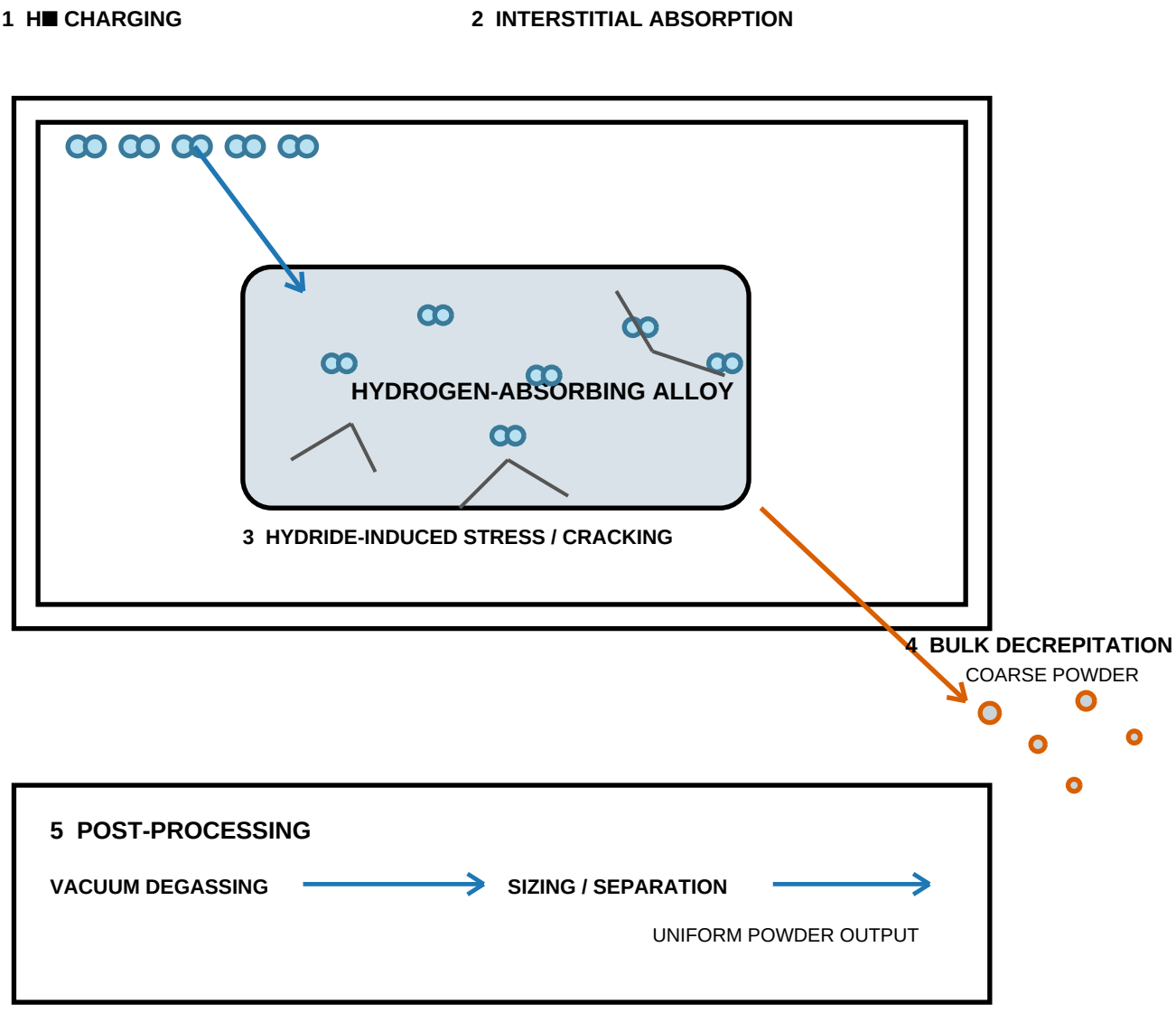


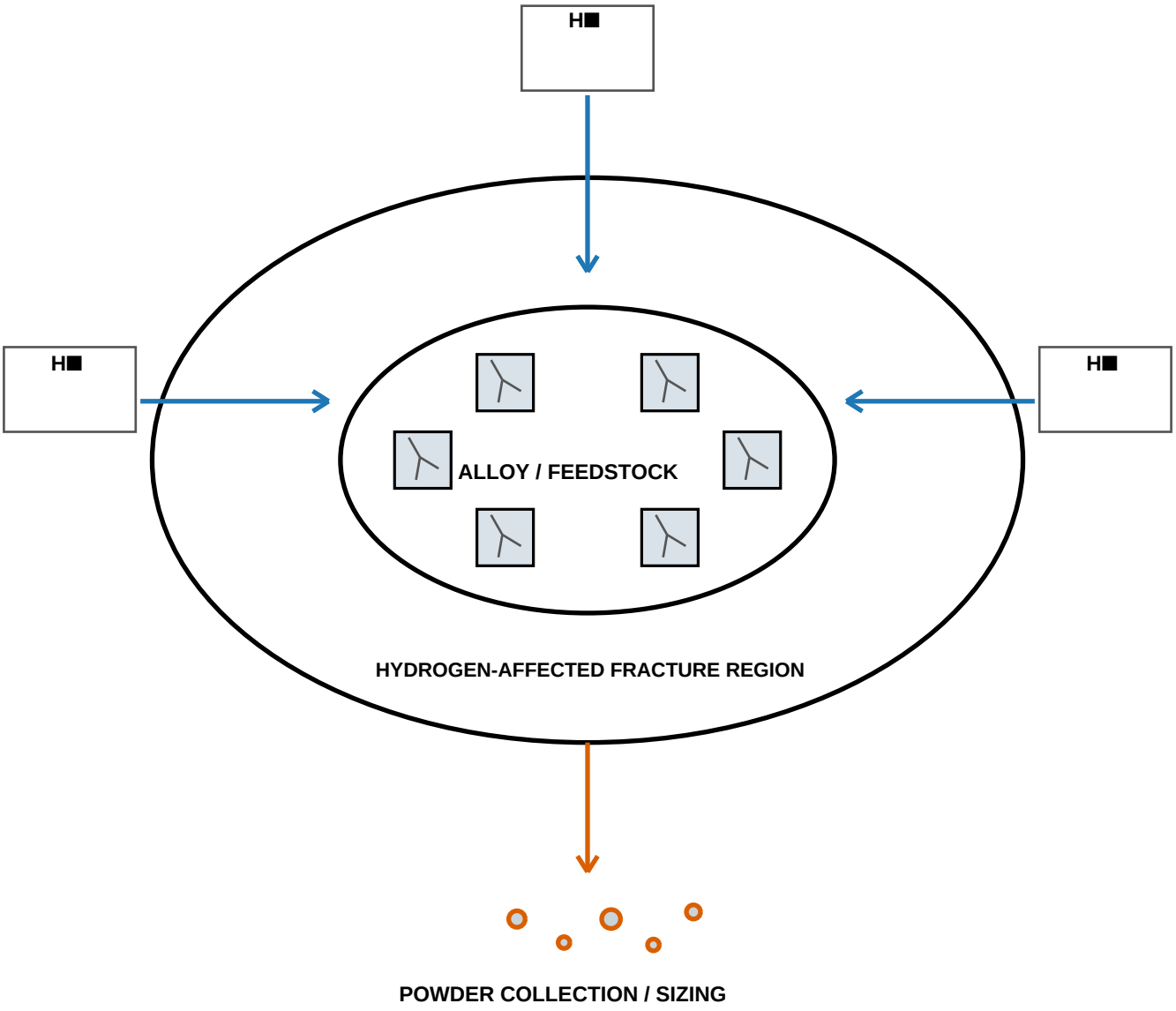
Perspective view showing the reduced functional sequence: hydrogen charging and absorption, hydrogen-induced cracking/decrepitation, followed by degassing and sizing.

The five detailed operations in the supplied process graphic are consolidated into three functional stages while retaining the original mechanisms and outputs.



H■ CHARGING → ABSORPTION → HYDRIDE STRESS → DECREPITATION → DEGASSING / SIZING

Side/cross-sectional view concentrating the original five boxes into one continuous working sequence. The key physical mechanism is hydrogen-assisted lattice expansion and cracking, followed by powder recovery.



REDUCED FUNCTIONAL STAGES

A. Hydrogen charging + interstitial absorption

B. Hydrogen-induced lattice stress + cracking + bulk decrepitation

C. Vacuum degassing + sizing / powder recovery

Top view showing the relative arrangement of hydrogen entry, alloy feedstock, fracture region and powder recovery, in the same compact three-figure style as the reference document.

This reduced diagram set is based on the supplied Advanced Hydrogen Decrepitation process graphic and reorganizes its five detailed steps into three functional stages without adding a new mechanism.